

STE-ANNE-DE-BELLEVUE, QC, Canada

WMO: 713770

Lat: 45.4272N

Lon: 73.9292W

Elev: 39

StdP: 100.86

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -23.6	(c) -20.8	(d) -28.8	(e) 0.3	(f) -22.8	(g) -26.1	(h) 0.3	(i) -20.4	(j) 10.5	(k) -1.5	(l) 9.5	(m) -4.6	(n) 3.4	(o) 250	(p) 0.536

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.4	30.1	21.9	28.5	21.0	27.1	20.2	23.5	27.9	22.5	26.6	21.6	25.3	4.1	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
22.1	16.8	26.0	21.2	15.9	24.8	20.2	15.0	24.0	70.2	28.0	66.4	26.6	63.1	25.2	28.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 8.7	(b) 7.7	(c) 6.9	(e) -27.0	(f) 32.5	(g) 2.9	(h) 1.3	(i) -29.0	(j) 33.4	(k) -30.7	(l) 34.2	(m) -32.3	(n) 34.9	(o) -34.4	(p) 35.9		
(4) 8.7	(5) 7.7	(6) 6.9	(e) -27.3	(f) 25.3	(g) 2.7	(h) 1.2	(i) -29.3	(j) 26.2	(k) -30.9	(l) 26.9	(m) -32.4	(n) 27.5	(o) -34.3	(p) 28.4		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	7.0	-9.1	-7.6	-2.0	6.1	13.7	18.6	21.2	20.1	16.0	9.2	2.1	-4.9
	DBStd	11.68	7.04	6.11	5.82	4.61	4.20	3.65	2.76	2.95	4.02	4.27	4.91	6.13
	HDD10.0	2372	591	493	374	133	12	0	0	0	2	68	238	461
	HDD18.3	4373	849	727	631	367	154	39	6	16	93	285	486	719
	CDD10.0	1287	0	0	1	16	126	259	346	312	181	43	2	0
	CDD18.3	246	0	0	0	1	10	49	94	70	22	1	0	0
	CDH23.3	1758	0	0	1	7	104	379	651	454	158	5	0	0
	CDH26.7	401	0	0	0	1	22	94	154	96	34	0	0	0
Wind	WSAvg	3.3	3.6	3.6	3.6	3.7	3.4	3.0	2.9	2.6	2.7	3.2	3.5	3.5
Precipitation	PrecAvg	1001	73	60	65	87	86	96	96	89	88	97	80	88
	PrecMax	1315	127	123	116	150	171	160	151	192	173	204	140	160
	PrecMin	821	31	25	20	16	35	55	47	24	29	30	21	24
	PrecStd	118	25	21	22	38	32	34	28	36	32	45	26	32
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	8.1	7.3	15.6	24.0	29.2	31.6	32.1	31.3	30.3	24.0	16.6	10.5
		MCWB	7.3	4.9	10.3	15.7	19.4	22.7	23.5	22.7	22.0	17.9	13.1	9.2
	2%	DB	5.1	4.6	10.5	19.5	26.2	29.3	29.9	29.0	27.1	20.8	13.7	7.1
		MCWB	3.7	2.8	6.5	11.8	17.5	20.9	22.2	21.5	20.7	16.0	11.4	5.9
	5%	DB	3.0	2.9	7.5	16.3	23.7	27.3	28.3	27.4	25.0	18.0	11.4	4.6
		MCWB	1.9	1.6	4.3	9.9	15.5	19.8	21.4	20.6	19.4	14.1	9.1	3.5
	10%	DB	0.9	1.4	5.4	13.5	21.3	25.5	26.9	25.9	22.8	15.7	9.1	2.7
		MCWB	0.0	0.3	2.8	8.3	14.8	18.7	20.4	20.1	18.4	12.6	7.0	1.6
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	7.4	5.6	12.0	16.6	21.1	24.2	25.3	24.3	23.3	19.2	14.0	9.4
		MCDB	7.9	6.7	14.3	22.4	26.8	29.3	30.6	29.1	27.9	22.3	15.4	10.6
	2%	WB	4.1	3.2	7.2	13.3	19.2	22.6	23.8	22.9	21.7	16.6	11.6	5.9
		MCDB	4.8	4.3	10.0	17.8	23.6	27.1	27.8	27.3	25.3	19.7	13.5	7.0
	5%	WB	1.9	1.7	4.7	10.8	17.5	21.4	22.6	21.9	20.4	14.7	9.4	3.5
		MCDB	2.9	2.7	7.0	15.0	21.6	25.3	26.5	25.4	23.9	17.5	11.2	4.4
	10%	WB	0.2	0.4	3.1	8.9	15.8	20.1	21.7	21.0	19.1	13.0	7.1	1.7
		MCDB	1.0	1.4	5.0	12.8	19.6	24.0	25.4	24.3	22.2	15.4	8.8	2.5
Mean Daily Temperature Range	5% DB	MDBR	8.1	8.6	8.3	9.3	10.3	9.8	9.4	9.7	10.0	8.3	6.9	6.7
		MCDBR	9.5	8.6	10.5	14.6	14.1	12.3	11.1	11.3	12.2	11.9	10.6	7.8
		MCWBR	9.1	7.9	7.6	8.7	7.1	6.0	5.5	5.4	6.6	7.5	8.6	7.3
	5% WB	MCDBR	9.0	8.3	9.8	12.7	11.5	10.2	9.6	9.6	10.9	10.7	10.2	7.6
		MCWBR	8.8	7.9	7.8	8.5	7.0	6.0	5.6	5.4	6.6	7.5	8.8	7.3
Clear-Sky Solar Irradiance	taub	0.261	0.289	0.329	0.362	0.387	0.443	0.430	0.414	0.374	0.338	0.307	0.289	
	taud	2.351	2.267	2.306	2.364	2.351	2.198	2.245	2.284	2.383	2.490	2.489	2.325	
	Ebn at Noon	858	891	899	897	881	829	836	835	842	831	798	770	
	Edh at Noon	74	99	111	115	121	141	133	123	103	80	67	70	
All-Sky Solar Radiation	RadAvg	1.43	2.38	3.64	4.41	5.14	5.62	5.68	5.03	3.97	2.31	1.41	1.01	
	RadStd	0.16	0.24	0.30	0.49	0.50	0.43	0.44	0.23	0.33	0.21	0.17	0.08	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (44 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page